VS Code • C++17 • task01.cpp • Terminal

```
$ g++ -std=c++17 -Wall -Wextra task01.cpp -o task01
$ ./task01
OUTPUT
Card object created
Can travel with Rs 20: true

--- Source code ---
 1  #include <iostream>
 2  using namespace std;
 3  class SmartCard {
 4      long long balance;
 5  public:
 6      SmartCard(long long b): balance(b) {}
 7      bool canTravel(long long fare) const { return balance-fare >= 5000; }
 8  };
 9  int main(){
10      SmartCard card(10000);
11      cout << "Card object created" << endl;
12      cout << "Can travel with Rs 20: " << boolalpha << card.canTravel(2000) << endl;
```

VS Code • C++17 • task08.cpp • Terminal

```
$ g++ -std=c++17 -Wall -Wextra task08.cpp -o task08
$ ./task08
OUTPUT
Hour accepted: 17
Fare: Rs 36.00

--- Source code ---
 1  #include <iomanip>
 2  #include <iostream>
 3  #include <sstream>
 4  using namespace std;
 5  int main(){
 6      string input="17"; int hour; istringstream in(input);
 7      if(!(in>>hour)) cout<<"Invalid hour"<<endl;
 8      else cout<<"Hour accepted: "<<hour<<endl;
 9      long long paise=3600;
10      cout<<"Fare: Rs "<<paise/100<<'.'<<setw(2)<<setfill('0')<<paise%100<<endl;
```

VS Code • C++17 • task07.cpp • Terminal

```
$ g++ -std=c++17 -Wall -Wextra task07.cpp -o task07
$ ./task07
OUTPUT
Procedural remaining: Rs 70
OOP deduction: true
OOP balance: Rs 70

--- Source code ---
 1  #include <iostream>
 2  using namespace std;
 3  long long proceduralFare(long long balance,long long fare){return balance-fare>=5000?balance-fare:-1;}
 4  class SmartCard{long long balance;public:SmartCard(long long b):balance(b){} bool canTravel(long long f)const{return balance-f>=5000
```

VS Code • C++17 • task03.cpp • Terminal

```
$ g++ -std=c++17 -Wall -Wextra task03.cpp -o task03
$ ./task03
OUTPUT
farePaise type: long long
sizeof(long long): 8 bytes
Network total: 9000000000 paise
INT_MAX: 2147483647

--- Source code ---
 1  #include <iostream>
 2  #include <climits>
 3  using namespace std;
 4  int main(){
 5      long long farePaise=3000;
 6      long long journeys=3000000;
 7      cout << "farePaise type: long long" << endl;
 8      cout << "sizeof(long long): " << sizeof(long long) << " bytes" << endl;
 9      cout << "Network total: " << farePaise*journeys << " paise" << endl;
10      cout << "INT MAX: " << INT MAX << endl;
```

VS Code • C++17 • task05.cpp • Terminal

```
$ g++ -std=c++17 -Wall -Wextra task05.cpp -o task05
$ ./task05
OUTPUT
11 16
9 6 4
6 4
3 1 -3 -1
18

--- Source code ---
 1  #include <iostream>
 2  using namespace std;
 3  int main(){
 4      int x=5,y=3;
 5      cout << x+y*2 << " " << (x+y)*2 << endl;
 6      int z=x++ + ++y;
 7      cout << z << " " << x << " " << y << endl;
 8      cout << (x>y?x:y) << " " << x%4*2 << endl;
 9      cout << 10/3 << " " << 10%3 << " " << -10/3 << " " << -10%3 << endl;
10      int k=0; for(int i=0;i<10;i+=3) k+=i; cout << k << endl;
```

VS Code • C++17 • task04.cpp • Terminal

```
$ g++ -std=c++17 -Wall -Wextra task04.cpp -o task04
$ ./task04
OUTPUT
3 3.5
C 68
1 true
s >= u

--- Source code ---
 1  #include <iostream>
 2  using namespace std;
 3  int main(){
 4      int a=7,b=2;
 5      double c=a/b;
 6      double d=static_cast<double>(a)/b;
 7      cout << c << " " << d << endl;
 8      char ch='A'+2;
 9      cout << ch << " " << ch+1 << endl;
10      bool flag=5;
11      cout << flag << " " << boolalpha << flag << endl;
12      unsigned int u=1; int s=-1;
13      cout << (s<u ? "s < u" : "s >= u") << endl;
```

VS Code • C++17 • task10.cpp • Terminal

```
$ g++ -std=c++17 -Wall -Wextra task10.cpp -o task10
$ ./task10
OUTPUT
20000 journeys: integer=60000000 paise, double=Rs 600000.00
500000 journeys: integer=1500000000 paise, double=Rs 15000000.00
3000000 journeys: integer=9000000000 paise, double=Rs 90000000.00
int total=410065408 (overflow)
long long total=9000000000 paise

--- Source code ---
 1  #include <iostream>
 2  #include <iomanip>
 3  using namespace std;
 4  int main(){
 5   long long journeys[]={20000,500000,3000000};
 6   for(long long n:journeys){long long paise=n*3000; double rupees=n*30.0; cout<<n<<" journeys: integer="<<paise<<" paise, double=Rs "
 7   int x=3000000*3000; long long y=3000000LL*3000; cout<<"int total="<<x<<" (overflow)"<<endl; cout<<"long long total="<<y<<" paise"<<
```

VS Code • C++17 • task09.cpp • Terminal

```
$ g++ -std=c++17 -Wall -Wextra task09.cpp -o task09
$ ./task09
OUTPUT
zone=1 peak=0 expected=1000 actual=1000 PASS
zone=2 peak=0 expected=2000 actual=2000 PASS
zone=3 peak=0 expected=3000 actual=3000 PASS
zone=1 peak=1 expected=1200 actual=1200 PASS
zone=3 peak=1 expected=3600 actual=3600 PASS
Passed: 5/5

--- Source code ---
 1  #include <iostream>
 2  using namespace std;
 3  long long fare(int zone,bool peak){long long b=zone==1?1000:zone==2?2000:zone==3?3000:0;return peak?b+b*20/100:b;}
 4  int main(){
 5   int passed=0; struct T{int z;bool p;long long e;}; T t[]={{1,0,1000},{2,0,2000},{3,0,3000},{1,1,1200},{3,1,3600}};
 6   for(auto &x:t){auto a=fare(x.z,x.p); cout<<"zone="<<x.z<<" peak="<<x.p<<" expected="<<x.e<<" actual="<<a<<" "<<(a==x.e?"PASS":"FAIL
 7   cout<<"Passed: "<<passed<<"/5"<<endl;
```

VS Code • C++17 • task06.cpp • Terminal

```
$ g++ -std=c++17 -Wall -Wextra task06.cpp -o task06
$ ./task06
OUTPUT
Fare Rs 10.00 balance Rs 90
Fare Rs 36.00 balance Rs 54
Denied (low balance) balance Rs 54

--- Source code ---
 1  #include <iomanip>
 2  #include <iostream>
 3  using namespace std;
 4  enum class Zone { Inner, Middle, Outer };
 5  constexpr int BASE_PAISE[]={1000,2000,3000};
 6  constexpr int PEAK_SURCHARGE_PERCENT=20;
 7  constexpr long long MIN_BALANCE_PAISE=5000;
 8  long long fare(Zone z,bool peak){
 9   int base=0;
10   switch(z){case Zone::Inner:base=BASE_PAISE[0];break;case Zone::Middle:base=BASE_PAISE[1];break;case Zone::Outer:base=BASE_PAISE[2];
11   return peak ? base + base*PEAK_SURCHARGE_PERCENT/100 : base;
12  }
13  bool canTravel(long long balance,long long f){return balance-f>=MIN_BALANCE_PAISE;}
14  int main(){long long balance=10000; struct Trip{Zone zone;bool peak;}; Trip trips[]={{Zone::Inner,false},{Zone::Outer,true},{Zone::M
```

VS Code • C++17 • task02.cpp • Terminal

```
$ g++ -std=c++17 -Wall -Wextra task02.cpp -o task02
$ ./task02
OUTPUT
Outer fare = Rs 30.0

--- Source code ---
 1  #include <iostream>
 2  using namespace std;
 3  enum class Zone { Inner, Middle, Outer };
 4  constexpr int INNER=1000, MIDDLE=2000, OUTER=3000;
 5  int main(){
 6      Zone z=Zone::Outer;
 7      int fare = (z==Zone::Inner?INNER:(z==Zone::Middle?MIDDLE:OUTER));
 8      cout << "Outer fare = Rs " << fare/100 << "." << fare%100 << endl;
```

VS Code • C++17 • task01.cpp • Terminal

```
$ g++ -std=c++17 -Wall -Wextra task01.cpp -o task01
$ ./task01
OUTPUT
Card object created
Can travel with Rs 20: true

--- Source code ---
 1  #include <iostream>
 2  using namespace std;
 3  class SmartCard {
 4      long long balance;
 5  public:
 6      SmartCard(long long b): balance(b) {}
 7      bool canTravel(long long fare) const { return balance-fare >= 5000; }
 8  };
 9  int main(){
10      SmartCard card(10000);
11      cout << "Card object created" << endl;
12      cout << "Can travel with Rs 20: " << boolalpha << card.canTravel(2000) << endl;
```